

Consumer Behavior Analytics Report

Executive Summary

This project presents an end-to-end Consumer Behavior Analytics solution developed to analyze retail transaction data and convert it into actionable business intelligence. The objective of the analysis was to understand revenue drivers, consumer purchasing behavior, product performance, and operational preferences to support strategic decision-making.

The analysis is based on **3,900 transaction-level records**, generating a total revenue of **\$233,080** with an **average purchase value of \$59.76** and an **average customer rating of 3.75**. Promotional offers were applied in **43% of total transactions**, indicating strong consumer responsiveness to discounts.

A fully interactive **Power BI application** was designed and deployed, featuring both **desktop and mobile-optimized dashboards**. The solution enables stakeholders to explore performance metrics, segment consumers, evaluate promotion and subscription effectiveness, and identify high-value transaction patterns.

1. Business Problem Statement

Retail organizations generate large volumes of transactional data daily; however, without structured analytics, it becomes difficult to translate this data into meaningful insights. Key business challenges include:

- Identifying primary revenue-driving customer segments
- Understanding product and category performance
- Measuring the impact of promotions and subscriptions
- Recognizing high-value and abnormal transaction behavior
- Enabling leadership with real-time, self-service analytics

This project addresses these challenges by building a centralized analytics dashboard that transforms raw transactional data into decision-ready insights.

2. Business Objectives

The primary objectives of this project were:

- Analyze overall revenue performance and spending distribution
- Understand customer demographics and purchasing behavior

- Identify top-performing products and categories
 - Evaluate promotion and subscription effectiveness
 - Assess operational preferences such as payment and shipping methods
 - Deliver a professional, enterprise-grade BI application
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3. Dataset Overview

Data Source

Retail consumer transaction dataset

Dataset Characteristics

- Total Transactions: **3,900**
- Total Customers: **3,900**
- Granularity: **Transaction-level (one record per purchase)**
- Final Attributes: **23 analytical fields**

Key Data Elements

- Customer demographics (age, gender, age group)
 - Product details (item, category, size, color)
 - Transaction values (purchase amount, units sold)
 - Operational attributes (payment method, shipping type)
 - Behavioral indicators (previous purchases, promotions, subscriptions)
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4. End-to-End Project Architecture

The project followed a complete analytics lifecycle:

Raw Data (CSV) → Data Cleaning & Validation (Python) → Feature Engineering → SQL-Based Business Analysis → Power BI Semantic Model → Power BI Service Deployment → Power BI App (Desktop & Mobile)

This ensured full traceability, scalability, and analytical integrity.

5. Data Preparation & Feature Engineering

Data Cleaning Activities

- Validation of numerical ranges for age, ratings, and purchase amount
- Standardization of categorical values
- Handling of missing and inconsistent entries
- Removal of redundant fields

Feature Engineering

Business-driven features were created to enhance analytical depth:

- **Age Group Classification** (Teen, Young Adult, Adult, Senior)
- **Promotion Exposure Flag**
- **High-Value Transaction Flag**
- **Express Shipping Indicator**
- **Purchase Frequency Approximation**
- **Outlier Transaction Identification**

These features enabled behavioral and value-based analysis beyond basic reporting.

6. Dashboard Design Overview

Dashboard Name

Consumer Insights & Revenue Performance Dashboard

Design Principles

- Executive-first KPI layout
- Consistent navigation across pages
- Dark enterprise theme
- Interactive slicers for dynamic filtering
- Page-level storytelling
- Optimized mobile experience

The dashboard was designed to support both **strategic review** and **detailed exploration**.

7. Dashboard Pages & Analytical Insights

7.1 Executive Summary Page

Purpose: Provide a consolidated overview of business performance.

Key components include: - Total Revenue - Total Customers - Units Sold - Average Rating - Promotion Usage

Insights: - Revenue distribution shows balanced contribution across categories - Customer base demonstrates stable engagement levels - Promotions are actively utilized without reducing order value

(Dashboard Screenshot – Summary Page)



7.2 Revenue Analysis Page

Focus Areas: - Revenue by Category - Revenue by Season - Revenue by Gender - Revenue by Age Group - Subscription Impact

Key Findings: - Clothing category contributes the highest share of total revenue - Adult and Young Adult segments generate the majority of sales - Male customers contribute a larger revenue share - Subscribed customers demonstrate higher spending behavior

(Dashboard Screenshot – Revenue Analysis)

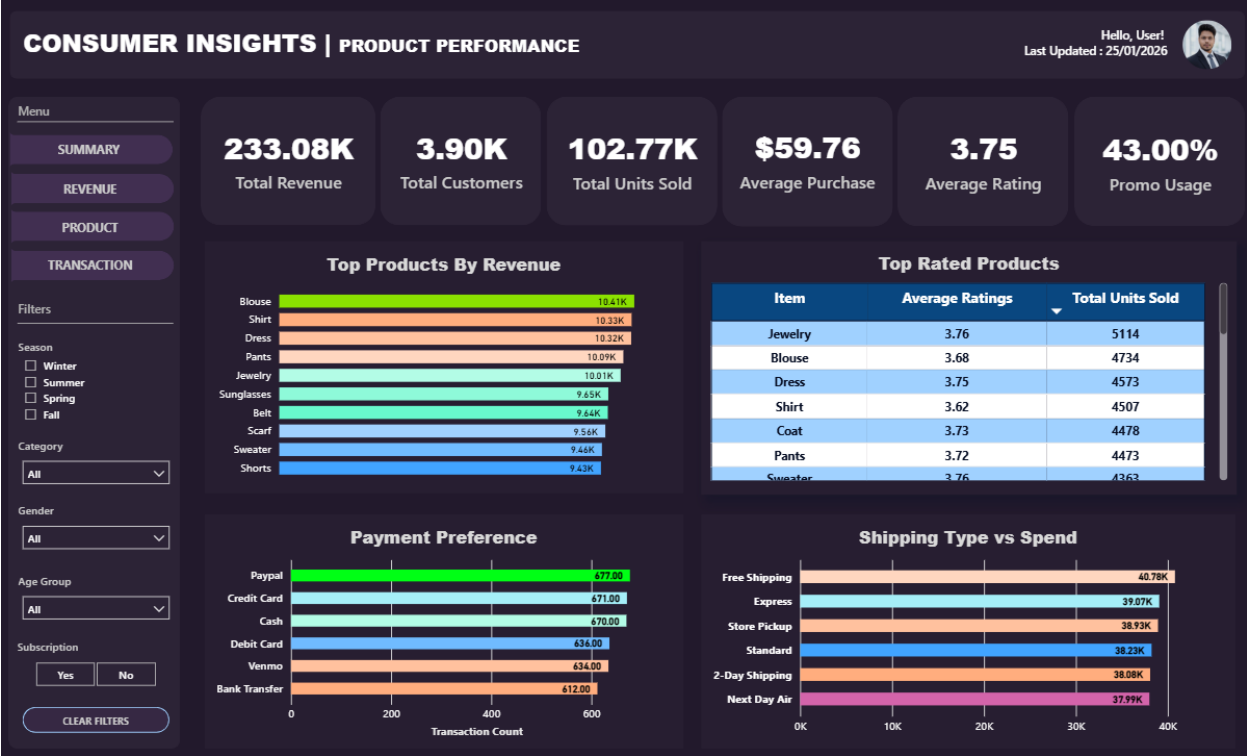


7.3 Product Performance Page

Focus Areas: - Top Products by Revenue - Product Ratings Analysis - Payment Method Preferences - Shipping Type vs Spend

Key Findings: - A small group of products contributes disproportionately to revenue - Average product ratings remain stable between 3.7–3.8 - Digital payment methods dominate transaction volume - Faster shipping options correlate with slightly higher spend

(Dashboard Screenshot – Product Performance)



7.4 Transactions Data Page

Purpose: Enable granular transaction-level analysis and validation.

Features: - Detailed transaction table - Multi-dimensional filtering - Drill-through support

This page supports operational review and data transparency.

(Dashboard Screenshot – Transactions Table)

CONSUMER INSIGHTS | TRANSACTIONS DATA

Hello, User!

Last Updated : 25/01/2026

Menu

SUMMARY

REVENUE

PRODUCT

TRANSACTION

Filters

Season

☐ Winter
 ☐ Summer
 ☐ Spring
 ☐ Fall

Category

All

Gender

All

Age Group

All

Subscription

CLEAR FILTERS

ID	Age	Gender	Product	Size	Colour	Category	Amount Paid	Payment Method	Shipping	Location	Rating	Previous Purchases	Subscribed
1	55	Male	Blouse	L	Gray	Clothing	53	Venmo	Express	Kentucky	3.10	14	Yes
2	19	Male	Sweater	L	Maroon	Clothing	64	Cash	Express	Maine	3.10	2	Yes
3	50	Male	Jeans	S	Maroon	Clothing	73	Credit Card	Free Shipping	Massachusetts	3.10	23	Yes
4	21	Male	Sandals	M	Maroon	Footwear	90	Paypal	Next Day Air	Rhode Island	3.50	49	Yes
5	45	Male	Blouse	M	Turquoise	Clothing	49	Paypal	Free Shipping	Oregon	2.70	31	Yes
6	46	Male	Sneakers	M	White	Footwear	20	Venmo	Standard	Wyoming	2.90	14	Yes
7	63	Male	Shirt	M	Gray	Clothing	85	Cash	Free Shipping	Montana	3.20	49	Yes
8	27	Male	Shorts	L	Charcoal	Clothing	34	Credit Card	Free Shipping	Louisiana	3.20	19	Yes
9	26	Male	Coat	L	Silver	Outerwear	97	Venmo	Express	West Virginia	2.60	8	Yes
10	57	Male	Handbag	M	Pink	Accessories	31	Cash	2-Day Shipping	Missouri	4.80	4	Yes
11	53	Male	Shoes	L	Purple	Footwear	34	Bank Transfer	Store Pickup	Arkansas	4.10	26	Yes
12	30	Male	Shorts	S	Olive	Clothing	68	Bank Transfer	Store Pickup	Hawaii	4.90	10	Yes
13	61	Male	Coat	M	Gold	Outerwear	72	Venmo	Express	Delaware	4.50	37	Yes
14	65	Male	Dress	M	Violet	Clothing	51	Paypal	Express	New Hampshire	4.70	31	Yes
15	64	Male	Coat	L	Teal	Outerwear	53	Debit Card	Free Shipping	New York	4.70	34	Yes
16	64	Male	Skirt	M	Teal	Clothing	81	Paypal	Store Pickup	Rhode Island	2.80	8	Yes
17	25	Male	Sunglasses	S	Gray	Accessories	36	Debit Card	Next Day Air	Alabama	4.10	44	Yes
18	53	Male	Dress	XI	Lavender	Clothing	38	Venmo	2-Day Shipping	Mississippi	4.70	36	Yes
19	52	Male	Sweater	S	Black	Clothing	48	Cash	Free Shipping	Montana	4.60	17	Yes
20	65	Male	Pants	M	Green	Clothing	90	Debit Card	Standard	Rhode Island	3.30	45	Yes
21	21	Male	Pants	M	Black	Clothing	51	Cash	Express	Louisiana	2.80	50	Yes
22	31	Male	Pants	M	Charcoal	Clothing	62	Debit Card	Store Pickup	North Carolina	4.10	22	Yes
23	56	Male	Pants	M	Peach	Clothing	37	Debit Card	Store Pickup	California	3.20	32	Yes
24	31	Male	Pants	XI	White	Clothing	88	Credit Card	Express	Oklahoma	4.40	40	Yes
25	18	Male	Jacket	M	Green	Outerwear	22	Debit Card	Store Pickup	Florida	2.90	16	Yes
26	18	Male	Hoodie	M	Silver	Clothing	25	Paypal	Express	Texas	3.60	14	Yes
27	38	Male	Jewelry	M	Red	Accessories	20	Credit Card	Next Day Air	Nevada	3.60	13	Yes
28	56	Male	Shorts	L	Cyan	Clothing	56	Bank Transfer	Next Day Air	Kentucky	5.00	7	Yes
29	54	Male	Handbag	M	Gray	Accessories	94	Paypal	Free Shipping	North Carolina	4.40	41	Yes
30	31	Male	Dress	S	Black	Clothing	48	Credit Card	Store Pickup	Wyoming	4.10	14	Yes
31	57	Male	Jewelry	L	Black	Accessories	31	Credit Card	Standard	North Carolina	4.70	16	Yes

8. Mobile Dashboard Experience

A dedicated mobile layout was developed to support executive and on-the-go consumption.

Mobile Design Highlights: - KPI-first vertical layout - Simplified visuals - Optimized slicers - Consistent branding

The mobile dashboard allows leadership to access insights anytime via the Power BI mobile application.

(Mobile Dashboard Screenshots)

- Menu
- SUMMARY

REVENUE

PRODUCT

TRANSACTION

Filters

Season

☐ Winter

☐ Summer

☐ Spring

☐ Fall

Category

All

Gender

All

Age Group

All

Subscription

All

CLEAR FILTERS

233.08K	3.90K
Total Revenue	Total Customers
102.77K	3.75
Total Units Sold	Average Rating
43.00%	
Promo Usage	



Menu

SUMMARY

REVENUE

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Category

All

Gender

All

Age Group

All

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All

CLEAR FILTERS

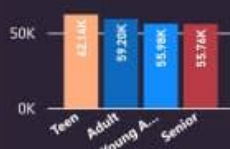
233.08K	3.90K
Total Revenue	Total Customers
102.77K	3.75
Total Units Sold	Average Rating
43.00%	\$59.76
Promo Usage	Average Purchase

Revenue By Gender



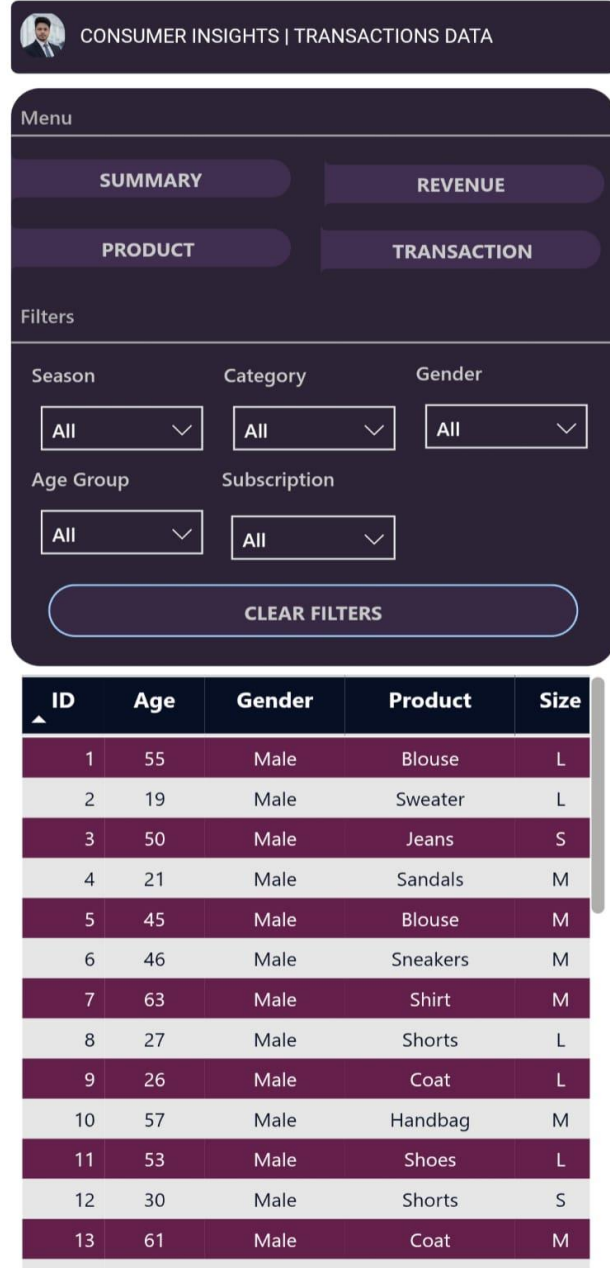
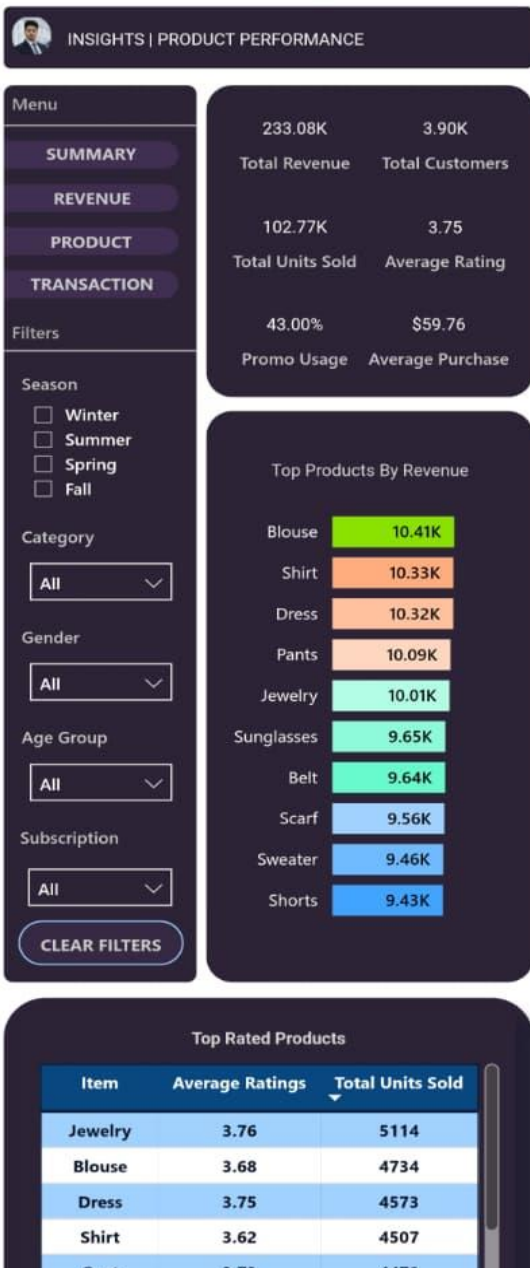
Gender	Revenue
Male	137.89K
Female	75.19K

Revenue By Age Group



Age Group	Revenue
Teen	42.10K
Adult	59.20K
Young Adult	55.98K
Senior	55.78K





9. Key Business Insights

- Clothing and Accessories are the strongest revenue-generating categories
- Adult and Young Adult customers represent the highest value segments
- Promotions drive engagement without lowering average order value
- Subscription customers exhibit stronger purchasing behavior
- Seasonal revenue remains balanced, with category-level variation
- Express shipping and digital payments align with higher spend patterns

10. Business Recommendations

Marketing Strategy

- Implement targeted promotions instead of blanket discounts
- Focus campaigns on high-spend demographic segments

Product Strategy

- Prioritize inventory planning for top-performing categories
- Use high-rated products for bundling and cross-selling

Customer Engagement

- Convert repeat buyers into subscribers through loyalty incentives
- Personalize offers using behavioral indicators

Operational Optimization

- Promote express shipping as a premium service
 - Strengthen partnerships with preferred digital payment platforms
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11. Tools & Technologies Used

- Python (Pandas, NumPy)
 - SQL (Business Analysis Layer)
 - Power BI Desktop
 - Power BI Service
 - Power BI App
 - Gamma AI (Presentation)
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12. Limitations & Assumptions

- Dataset does not include transaction timestamps
- Purchase frequency values are approximated
- Review ratings are subjective

These limitations were addressed through feature engineering and analytical assumptions.

13. Conclusion

This project demonstrates the complete analytics lifecycle—from raw data processing to enterprise-level business intelligence delivery.

By integrating data engineering, SQL analytics, and Power BI visualization, the solution provides decision-makers with clear visibility into consumer behavior, revenue drivers, and strategic opportunities.

The Consumer Behavior Analytics Dashboard serves as a scalable foundation for data-driven retail decision-making and reflects industry-standard BI development practices.